

FI360 TYRE MODULE

UI/UX & SCREEN SPECIFICATION

Fleet Intelligence 360 (FI360) — Version 1.0
Fleet Associates Africa Ltd (FAA)
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Document Control

Field	Value
Document ID	FI360-UX-TYR-007
Version	1.0
Status	Baseline / Product Design Specification
Parent	FI360 Tyre SRS v1.0; Database & Data Dictionary v1.0; API & Integration Contract v1.0
Core Foundation	FI360 Core Architecture & Common Services Specification v1.0
Module	FI360 Tyre Management

1. Purpose

This document defines the user experience, navigation, screen architecture, field behaviour, workflows, responsive design and interaction standards for FI360 Tyre. It is the bridge between the approved requirements, data and API documents and the actual wireframes, prototypes and software screens.

The module must remain independently usable while becoming richer when Workshop, Fleet, Telematics or Analytics are later connected. The API contract supports standalone operation and additive integrations.

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2. UX Principles

- Technician-first: routine transactions require minimal clicks and typing.
- Manager-ready: KPIs and exceptions are visible immediately.
- Search-first: Tyre ID, serial, barcode/QR and Asset ID are fast search keys.
- Progressive disclosure: show essential information first and advanced detail on demand.
- Safe transactions: material actions require clear confirmation and reasons.
- History-first: the complete tyre lifecycle remains traceable.
- Mobile capable: inspection and field work must work on tablets/phones.
- Accessible: status is never communicated by colour alone.
- Consistent: reuse FI360 Core navigation, identity and common components.

3. Target Users

User	Main Needs
Tyre Technician	Scan, inspect, fit, remove, rotate
Tyre Supervisor	Daily operations, approvals, exceptions
Stores Officer	Receive, store, move and issue tyres
Tyre Manager	Lifecycle, performance, cost and KPIs
Fleet Manager	Vehicle/asset tyre visibility
Finance/Analyst	Cost and lifecycle reporting

Administrator	Configuration and permissions
Auditor	Read-only traceability

4. Application Shell

Element	Behaviour
Top bar	FI360 identity, module, tenant/site, notifications, help, user menu
Navigation	Dashboard, Tyres, Operations, Lifecycle, Analytics, Reports, Administration
Global search	Tyre ID, serial, barcode/QR, Asset ID
Breadcrumbs	Show current location
Page header	Title, context, status and primary action
Content	Cards, tables, forms, timelines and charts
Notifications	Success, warning, error and information
Help	Contextual guidance

5. Primary Navigation

Dashboard → Tyres → Operations → Lifecycle → Analytics → Reports → Administration

Core navigation

Area	Screens
Dashboard	Tyre Overview, Exceptions, KPIs
Tyres	Register, Search, Details
Operations	Receive, Stock, Fitment, Removal, Inspection, Rotation
Lifecycle	Repair, Retread, Retirement, Disposition
Analytics	Performance, Wear, Cost, Supplier/Brand
Reports	Standard Reports, Exports
Administration	Configuration, Thresholds, Positions, Permissions

6. Dashboard

Widget	Content
Total Tyres	Count by lifecycle status
In Service	Currently fitted/active tyres
In Stock	Available tyres by location
Inspection Compliance	Due versus completed
Low Tread	Tyres requiring attention
Pressure Exceptions	Out-of-range readings
Premature Removals	Trend and count
Cost/km	Configured lifecycle cost metric
Lifecycle Cost	Acquisition + repair + retread + configured costs
Alerts	Prioritized exceptions

Every KPI should be clickable into its underlying filtered records. Widgets must respect tenant, site and role permissions.

7. Tyre Register

Control	Behaviour
Search	Tyre ID, serial, barcode/QR, brand/model
Status	Multi-select lifecycle status

Specification	Size, brand and model filters
Location	Site/store/location
Asset	Current FI360 Asset ID
Date	Registration/fitment/inspection date depending view
Alert	Normal/warning/critical
Export	Permission-controlled export

- Use server-side pagination.
- Preserve filters during navigation where practical.
- Support saved operational views.
- Show current status, location, asset/position and latest inspection summary.

8. Tyre Details

Section	Content
Identity	Tyre ID, serial, brand, model, size, manufacture data
Current State	Status, location, current fitment
Condition	Latest inspection, tread and pressure
Asset Context	Asset, position and usage
Timeline	All lifecycle events
Cost	Acquisition, repair, retread and total lifecycle cost
Alerts	Open exceptions and actions
Documents	Authorized photos/documents
Audit	Material actions, permission controlled

A user should understand the complete history of a tyre without opening multiple unrelated screens.

9. Tyre Registration

Scan / Enter → Validate → Technical Data → Inspect → Register → Activate

Registration workflow

Field	Rule
Tyre ID	System-generated
Serial Number	Configurable uniqueness rule
Brand/Model	Controlled references
Size	Required controlled specification
Manufacture Date	Where available
Supplier/Receipt	Where applicable
Initial Location	Required
Condition	Required
Evidence	Optional/configurable

- Support barcode/QR scanning.
- Validate before activation.
- Prevent duplicates according to policy.
- Provide review/confirmation before final posting where required.
- Open Tyre Details after successful registration.

10. Fitment

Select Tyre → Select Asset → Select Position → Validate → Confirm → Post

Fitment workflow

Step	Behaviour
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Tyre	Only eligible tyres can be selected
Asset	Use canonical FI360 Asset ID when Fleet is installed
Position	Show valid positions only
Validation	Check specification, active fitment and business rules
Usage	Capture mileage/hours, date/time and technician
Confirmation	Show complete before/after summary
Completion	Show success, audit and event reference

The API contract requires protected, versioned APIs and states that a Tyre-only customer does not need another module for operation. [filecite](#)[turn1file0](#)[L43-L53](#)

11. Removal

Field	Rule
Tyre / Asset / Position	Display current relationship
Date/time	Default current; controlled edit
Mileage/hours	Capture where available
Removal reason	Required
Condition	Required
Tread	Required where policy applies
Destination	Stock, repair, retread, retired or scrap
Notes/evidence	Configurable

Irreversible dispositions must use a strong confirmation step and reason capture.

12. Inspection

Open → Measure → Assess → Save → Alert → Action

Inspection workflow

Section	Fields
Context	Tyre, Asset, position, type, date/time, inspector
Tread	Measurement points, depth and unit
Pressure	Measured pressure, unit and tolerance context
Visual condition	Configured inspection checks
Overall condition	Controlled rating
Recommendation	Continue, monitor, repair, retread or remove
Evidence	Photos and notes

- Validate plausible measurement ranges.
- Display target/tolerance context.
- Never rely on colour alone.
- Keep previous inspections immutable.

13. Stock & Location

Feature	Behaviour
Stock summary	Counts by site, location, size, brand and status
Location tree	Site → store → bay/bin
Move	Controlled movement transaction
Issue	Controlled issue to operation/fitment
Ageing	Age bands for stored tyres
Reconciliation	Authorized physical/system reconciliation
Scan mode	Fast barcode/QR workflows

14. Rotation

- Show current positions.
- Select tyres to rotate.
- Display before/after positions.
- Capture mileage/hours and reason.
- Validate target positions.
- Show confirmation summary.
- Post one controlled transaction.

15. Repair & Retread

Screen	Core Fields
Repair	Tyre, repair type, technician/supplier, date, cost, result, notes, evidence
Retread	Casing, current cycle, new cycle, supplier, send/return dates, cost, result
Retirement	Tyre, reason, date, residual/disposition information
Scrap	Tyre, reason, approval, date, destination/evidence

16. Lifecycle Timeline

Event	Displayed Information
Registered	Date, actor
Received	Supplier, location, reference
Fitted	Asset, position, mileage/hours
Inspected	Condition, tread, pressure
Rotated	Old/new position
Removed	Reason, readings, destination
Repaired	Type, cost, result
Retreaded	Cycle, supplier, result
Retired/Scrapped	Final disposition

17. Alerts

Severity	UI Treatment
Critical	Prominent banner/card + action required + text indicator
Warning	Visible alert + due/action date
Information	Non-blocking notification
Resolved	Resolution status and history

- Explain what happened, why it matters and recommended action.
- Allow assignment/acknowledgement/resolution according to permissions.
- Never silently delete alert history.
- Link directly to affected tyre/transaction.

18. Reports

Report	Filters
Tyre Register	Status, brand, model, size, location
Stock	Location, size, brand, ageing
Fitment History	Asset, position, date
Inspection	Date, condition, compliance

Tread Analysis	Tyre, asset, size, date
Pressure Compliance	Asset, date, exception
Repair/Retread	Supplier, type/cycle, date
Lifecycle Cost	Tyre, asset, date, cost type
Premature Removal	Reason, target life, date
Disposition	Reason, date

19. Form Standards

- Labels remain visible; do not rely on placeholders.
- Mark required fields consistently.
- Use inline validation.
- Do not lose valid user input after an error.
- Use controlled reference lists.
- Use correct numeric/date input controls.
- Display measurement units.
- Use confirmation summaries for material actions.

20. Mobile / Field UX

Open Task → Scan → Capture → Validate → Queue / Sync → Confirm

Field interaction

- Large touch targets.
- Scan-first physical tyre workflows.
- Minimal typing.
- Photo capture where supported.
- Visible connectivity/sync state.
- Offline queue can be introduced without changing the core transaction model.
- Inspection forms group measurements logically.

21. Standalone vs Integrated Experience

Capability	Tyre Only	Tyre + Other FI360 Modules
Tyre management	Complete	Complete
Asset context	Approved standalone reference	Canonical Fleet Asset
Workshop	Not required	Contextual links/actions
Mileage/hours	Manual/approved input	Fleet/Telematics integration
Analytics	Tyre KPIs	Cross-module analytics
Navigation	Tyre only	Entitlement-based modules

The FI360 development standard requires independent operation and integration readiness; optional integrations must not destroy core module operation. ☐filecite☐turn1file3☐L312-L330☐

22. Future Workshop Connection

Tyre Event	Possible Workshop UX
Failed inspection	Create/open related work order
Removal	Show associated maintenance activity
Repair	Link to Workshop work order
Replacement required	Create/associate maintenance action
Fitment	Show installation activity

These links appear only when Workshop is enabled, authorized and available.

23. Permissions

Permission	UI Effect
tyre.read	View
tyre.create	Registration
tyre.receive	Receiving
tyre.fit	Fitment
tyre.remove	Removal
tyre.inspect	Inspection
tyre.rotate	Rotation
tyre.repair	Repair
tyre.retread	Retread
tyre.cost	Cost information
tyre.approve	Approval controls
tyre.admin	Administration

UI hiding is not security. The API must independently enforce authorization.

24. Responsive Design

Device	Design Principle
Desktop	Full navigation, dashboards and detailed tables
Tablet	Responsive forms and reduced table columns
Mobile	Stacked cards, task-first screens and scan/capture workflows
Field device	Scan → capture → validate → confirm

25. Accessibility

- Keyboard-accessible controls.
- Visible focus states.
- Readable contrast.
- Status must not depend on colour alone.
- Text-based validation messages.
- Logical heading hierarchy.
- Accessible icon labels.
- Usable at browser zoom.

26. Loading, Empty, Error & Offline States

State	Required Behaviour
Loading	Skeleton/spinner without unnecessary blocking
Empty	Explain why and suggest next action
Success	Confirm completion and next action
Error	Explain problem and recovery path
No permission	Clear access message without data leakage
Offline	Show offline state and queued/sync status where supported
Integration pending	Separate local transaction success from integration status

27. Search Experience

- Exact Tyre ID and serial matches first.
- Barcode/QR scan where supported.
- Recent searches where useful.
- Show current status/location in results.
- Open Tyre Details directly.
- Asset searches show only authorized vehicle context.

28. Confirmation Rules

Action	Confirmation
Register	Review/confirm where configured
Fit	Always confirm tyre + asset + position
Remove	Always confirm reason + destination
Rotate	Confirm before/after positions
Repair	Confirm cost/result
Retread	Confirm cycle/supplier/cost
Retire	Confirm consequence
Scrap	Strong confirmation + reason/approval

29. Design System

- Reuse FI360 Core components and design tokens.
- Standardize buttons, forms, tables, badges, alerts and dialogs.
- Use a consistent icon set.
- Use consistent typography hierarchy.
- Use text plus visual status indicators.
- Tenant branding should be controlled, not hard-coded into the Tyre module.

30. Prototype Priority

Priority	Screens
P0	Dashboard, Register, Search, Details, Fitment, Inspection, Removal
P1	Stock, Rotation, Repair, Retread, Alerts, Reports
P2	Advanced Analytics, predictive views, advanced integrations

The first prototypes should validate the operational loop: register → store → fit → inspect → remove → lifecycle history.

31. UI/API/Data Traceability

UI	Primary Backend Source
Tyre Register	Tyre API + tyre master tables
Tyre Details	Tyre + history APIs
Fitment	Fitment API
Inspection	Inspection API
Stock	Movement/location APIs
Repair/Retread	Lifecycle APIs
Dashboard	Analytics/KPI APIs
Alerts	Alert service/API
Reports	Reporting APIs

The API contract states that request/response payloads map to the data dictionary, making the data model and API contract the controlling references for field names. [filecite\turn1file0\L43-L50](#)

32. UX Acceptance Criteria

- Technician can register or scan a tyre efficiently.
- Technician can complete fitment with clear validation.
- Inspection can be completed on tablet/mobile.
- Manager can identify critical exceptions from dashboard.
- User can trace a tyre from registration to current state.
- Tyre-only customer encounters no broken optional dependencies.
- Integrated actions appear only when enabled and authorized.
- Protected actions correspond to API permissions.
- Critical actions have appropriate confirmation.
- Desktop, tablet and mobile layouts remain usable.

33. Design Handoff Package

Artifact	Required Output
Design system	Components, tokens, states and usage
Wireframes	All P0 screens
High-fidelity UI	Approved visual designs
Prototype	Clickable core workflows
Component specs	States and interactions
API mapping	UI fields mapped to API contracts
Data mapping	UI fields mapped to data dictionary
UX test script	Task-based usability scenarios

34. Next Document

DOCUMENT 08 — FI360 TYRE MODULE DEVELOPER BACKLOG, TEST PLAN & TEST CASES should now convert the SRS, database/data dictionary, API contract and this UI/UX specification into executable Epics, Features, User Stories, acceptance criteria, technical tasks, unit/integration/system tests, security tests, performance tests and UAT scenarios.

The FI360 development standard identifies UI/UX as mandatory and places it before API/event implementation and testing in the module implementation sequence. [filecite\turn1file4\L278-L290](#) [filecite\turn1file3\L240-L250](#)

UX principle: make the right tyre action obvious, fast and traceable.